
ReTok-VC: Variational Token-Level Video Compression with Learned Reuse

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Abstract

Recent tokenized video codecs reduce per-frame token traffic with hand-designed routing rules: TVC transmits a deterministic checkerboard subset of tokens at every frame, while TivTok partitions tokens once per clip into a time-invariant set that is broadcast across frames and a time-variant set transmitted per frame. Both rules are content-agnostic by construction, and both confine these codecs to the ultra-low bitrate regime. We present ReTok-VC, a tokenized video codec in which the per-token, per-frame decision to reuse temporal context or to transmit a residual is treated as the variational posterior of a structured discrete latent-variable model. The corresponding evidence lower bound recovers the Lagrangian rate-distortion loss as its β -weighted form, and the reuse indicator is itself entropy-coded under a learned prior, contributing well below one bit per token in static regions. The image tokenizer is held frozen, and only three video-specific modules receive gradients (a reuse predictor, a residual vector quantizer, and a conditional entropy model), trained by a stabilized two-stage schedule for robust multi-rate operation. On UVG and HEVC Class B, ReTok-VC matches or exceeds the strongest tokenized-codec baselines in their reported ultra-low regime and extends operation across the full target bitrate range.

1 Introduction

Image tokenizers have closed much of the gap to conventional intra-frame codecs: TiTok reconstructs an image from as few as 32 discrete tokens [Yu et al., 2024b], and MAGVIT-v2 reports tokenizer reconstruction quality on par with VVC [Yu et al., 2024a]. Two recent works lift this discrete substrate to video. TVC introduces a dual-stream design in which a fixed checkerboard mask selects which tokens to transmit and a decoder-only transformer reconstructs the rest [Zhou et al., 2025]; TivTok decomposes a clip into a time-invariant token set that is broadcast across frames and a time-variant set transmitted per frame [Chen et al., 2026]. Both report strong reconstruction quality at sub-0.03 bpp and are the first tokenized video codecs reported in the literature.

The transmission rules adopted by these codecs are, however, content-agnostic by construction. TVC’s checkerboard mask is identical for every video and every frame. TivTok’s TIV/TV split is fixed by architectural design at the clip level. Neither rule can adapt the bit budget to local motion: a static background patch and a moving object in the same frame are coded under the same allocation. This rigidity both confines the methods to the ultra-low bitrate regime, where a coarse rule remains adequate, and prevents extension into the mid-bitrate range where learned continuous-latent codecs such as DCVC-FM currently dominate [Li et al., 2024].

We argue that the per-token, per-frame decision to reuse temporal context should not be a hand-designed rule but the variational posterior of a structured discrete latent-variable model. Under this view, the latent of a P-frame consists of two random variables per token slot: a binary reuse indicator

and, conditional on transmission, a discrete residual code produced by a multi-stage residual quantizer. The negative evidence lower bound of the model decomposes into a reconstruction term and the expected coding lengths of the two streams, recovering the Lagrangian rate-distortion loss as its β -weighted form [Ballé et al., 2018, Minnen et al., 2018]. The reuse indicator itself is entropy-coded under a learned prior conditioned on temporal context, so its bit cost is paid only where the temporal reference is informative and is amortized away wherever consecutive frames agree.

ReTok-VC realizes this framework on top of any frozen image tokenizer. Three video-specific modules are trained: a per-token Bernoulli reuse predictor, a residual vector quantizer that exploits the low entropy of token residuals after temporal reuse to refine transmitted tokens in feature space, and a conditional entropy model that supplies coding distributions for both streams. Training proceeds in two stages: a stabilized single-rate optimization that prevents the early reuse-rate collapse common to discrete-latent codecs, followed by a multi-rate fine-tune that improves operating-point robustness. Holding the tokenizer frozen is a deliberate design choice that preserves the pretrained reconstruction floor and yields a reuse-and- residual stack portable across substrates. We develop the trade-off in Sec. 3.2.

The paper makes the following contributions:

- **Variational reformulation of tokenized video coding.** We recast tokenized video compression as variational lossy coding over a structured discrete latent comprising the underlying token, the reuse indicator, and the residual code, and we show that the standard Lagrangian rate-distortion loss arises as the β -weighted negative ELBO of this model.
- **Learned, RD-optimal, per-token reuse with entropy-coded indicator.** We introduce a per-token, per-frame reuse mechanism whose Bernoulli posterior is itself entropy-coded under a temporal-context-conditioned prior. To our knowledge this is the first per-token, per-frame learned reuse decision derived from a unified rate-distortion objective in tokenized video compression.
- **Tokenizer-portable, wide-rate operation.** We demonstrate that the framework operates on top of any frozen image tokenizer without tokenizer-specific retraining, and that the resulting model family covers an order of magnitude wider bitrate range than prior tokenized codecs while matching or exceeding their reported ultra-low-bitrate quality on UVG and HEVC Class B.

2 Related Work

Traditional and learned video compression. Three conventional video codecs serve as the standard reference points for any learned method: HEVC [Sullivan et al., 2012], VVC [Bross et al., 2021], and the in-development ECM beyond-VVC exploration. They are highly engineered block-hybrid pipelines whose coding primitives are largely fixed and difficult to adapt at fine semantic granularity. On the learned side, DVC [Lu et al., 2019] pioneered the end-to-end neural video codec by replacing traditional codec modules with neural networks while retaining explicit motion estimation and residual coding. The subsequent DCVC family [Li et al., 2021, 2023, 2024, Jia et al., 2025] replaced this paradigm with conditional coding in a learned continuous feature space, with DCVC-FM introducing feature modulation and a learnable quantization scaler for wide-rate support and DCVC-RT pursuing real-time decoding under operational-cost constraints. A separate line of work models videos as coordinate-conditioned implicit neural representations: HiNeRV [Kwan et al., 2023] encodes videos hierarchically and PNVC [Gao et al., 2025] reparameterizes the implicit representation for practical compression. Both directions operate inside dense continuous latents or coordinate fields, where the bit budget for temporal redundancy is allocated implicitly through entropy modeling rather than through an explicit per-token decision.

Tokenizers for visual compression. Discrete visual tokenizers map continuous frames into low-cardinality symbol sequences. VQ-VAE [Van Den Oord et al., 2017] establishes the codebook-based discrete autoencoder. MAGVIT-v2 introduces lookup-free quantization to support large vocabularies in a shared image-and-video codebook [Yu et al., 2024a], and TiTok demonstrates that as few as 32 one-dimensional semantic tokens suffice to reconstruct an image at high fidelity [Yu et al., 2024b]. Finite scalar quantization replaces learned codebooks with deterministic per-axis discretization for training and deployment stability [Mentzer et al., 2024] and is adopted by Cosmos [NVIDIA et al., 2025] and VidTok [Tang et al., 2024] to provide uniform tokenization stacks for both images and

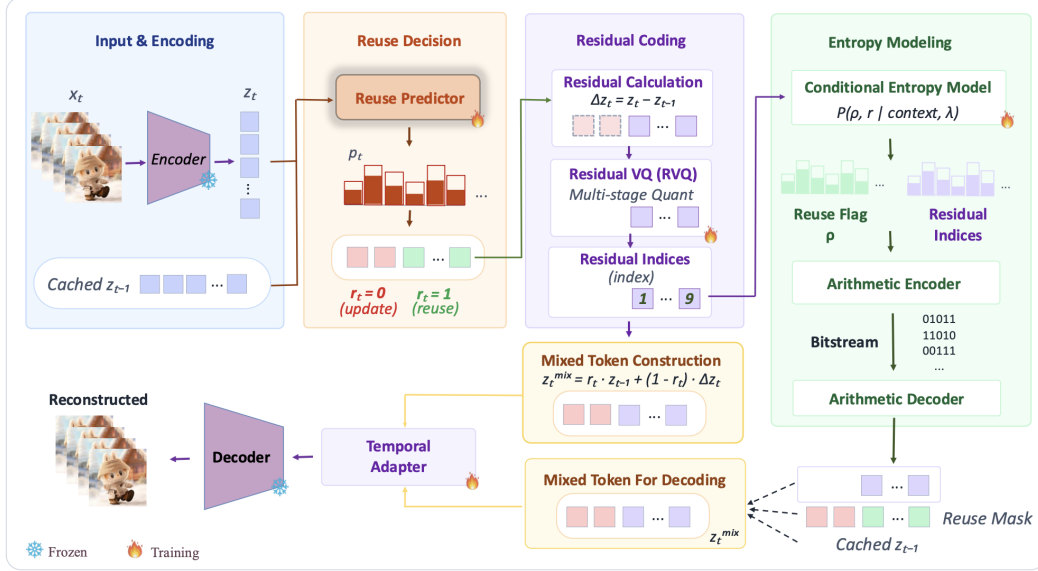


Figure 1: End-to-end pipeline of ReTok-VC. The frozen tokenizer encoder E produces per-frame token features z_t . The reuse predictor outputs the per-token indicator $\rho_i \in \{0, 1\}$, the residual vector quantizer encodes $\Delta z_t = z_t - z_{t-1}$ for tokens with $\rho_i = 0$, and the conditional entropy model supplies probabilities for the arithmetic coder over (ρ, r) . The receiver assembles z_t^{mix} from the cached z_{t-1} and the decoded residual, refines it with a temporal adapter, and produces $\hat{x}_t$ via the frozen decoder D . Snowflake denotes frozen modules, flame denotes modules trained under Eq. (8). Module-level inputs and outputs are specified in Sec. 3.

91 videos. Two implications follow from this lineage: discrete token indices are robust transmission units,
 92 and learned token spaces carry strong perceptual priors that approach next-generation codec quality
 93 without bespoke entropy modeling. However, image-origin tokenizers do not specify a temporal
 94 allocation policy, and applying them frame-by-frame to video underuses cross-frame redundancy
 95 unless an explicit reuse mechanism is introduced on top of them.

96 **Token-based video compression.** Two recent works define the token-centric video compression
 97 landscape. TVC [Zhou et al., 2025] introduces a dual-stream ultra-low-bitrate design in which
 98 discrete tokens are partially masked under a fixed checkerboard pattern and reconstructed by a
 99 decoder-only transformer with spatiotemporal token prediction, while a parallel continuous stream
 100 supplies complementary detail. TivTok [Chen et al., 2026] decomposes a video into time-invariant
 101 tokens that are broadcast across frames and time-variant tokens that are transmitted per frame. The
 102 decomposition is fixed by architectural design rather than driven by content, and the authors note
 103 that rigid predefined decomposition becomes brittle when scene dynamics shift. Both treat temporal
 104 redundancy as a first-class signal in token space, but their routing rules are content-agnostic, being
 105 either purely spatial (TVC) or globally factorized over a clip (TivTok). A parallel generative thread
 106 relies on diffusion priors for perceptual quality at extreme bitrates [Careil et al., 2024, Ma and
 107 Chen, 2025, ?], but allocates bits implicitly through generative sampling rather than through an
 108 explicit token-level decision. We develop the alternative in which the routing is learned at per-token,
 109 per-frame granularity under a unified rate-distortion objective.

110 3 Method

111 3.1 Problem setup and framework overview

112 Let $\{x_t\}_{t=1}^T$ be a video sequence with frames $x_t \in \mathbb{R}^{H_x \times W_x \times 3}$. A frozen pretrained image tokenizer
 113 $\mathcal{T} = (E, \mathcal{C}, D)$ maps each frame independently to a sequence of N discrete token indices $z_t =$
 114 $E(x_t) \in \{1, \dots, K\}^N$ over a codebook $\mathcal{C} \in \mathbb{R}^{K \times d}$ of size K , and the tokenizer’s own per-frame
 115 reconstruction is $D(\mathcal{C}[z_t])$. Throughout the model, the parameters of $E, \mathcal{C}$, and D are kept fixed; only

Table 1: Core notation used in ReTok-VC.

Symbol	Meaning	Symbol	Meaning
x_t	Frame at time t	K, K_r	Tokenizer / RVQ-stage codebook size
$z_t = \{z_i^t\}_{i=1}^N$	N token indices for x_t	Q	Number of RVQ stages
$\hat{z}_i^t$	Reconstructed token feature	W	Context window length
$\rho_i^t \in \{0, 1\}$	Reuse indicator (1: reuse)	f_ϕ, g_ψ	Reuse predictor, context summary
$r_i^t = (r_i^{t,1}, \dots, r_i^{t,Q})$	RVQ index sequence (when $\rho_i^t = 0$)	$h_\theta^p, h_{\theta,q}^r$	Entropy-model heads
c_i^t, c_t	Per-token / bundled temporal context	$\lambda, \lambda_{\text{mid}}$	RD coefficient (default $\lambda_{\text{mid}}=10^{-2}$)
$\mathcal{D}$	Reconstruction distortion	α_{lpiPs}	LPIPS weight (default 0.5)
$\mathcal{R}_\rho, \mathcal{R}_r$	Coding rates of ρ and r		

116 video-specific modules receive gradients. All symbols introduced in this section are summarized in
 117 Table 1.

118 We code one I-frame followed by $T - 1$ P-frames. The I-frame is coded by an off-the-shelf intra
 119 codec. The remainder of this section specifies the P-frame model. For each P-frame, every token slot
 120 $i \in \{1, \dots, N\}$ carries a binary reuse indicator $\rho_i^t \in \{0, 1\}$ together with, when $\rho_i^t = 0$, a residual
 121 symbol r_i^t . The reconstructed token feature is

$$\hat{z}_i^t = \mathcal{C}[\hat{z}_i^{t-1}] + (1 - \rho_i^t) \Delta_q(r_i^t), \quad (1)$$

122 where $\Delta_q(\cdot)$ is the residual decoder defined in Eq. (4). A reused token ($\rho_i^t = 1$) costs no residual
 123 bits and is identical to the previous reconstruction at the same slot. A transmitted token ($\rho_i^t = 0$) is
 124 refined by the residual stream. The frame is reconstructed as $\hat{x}_t = D(\hat{z}_t)$ with the frozen D .

125 Three core modules are trained: a per-token reuse predictor f_ϕ (Sec. 3.2), a Q -stage residual vector
 126 quantizer (Sec. 3.2), and a conditional entropy model that supplies coding distributions for both reuse
 127 indicators and residual symbols (Sec. 3.4). A small trainable temporal adapter additionally refines
 128 the assembled feature sequence z_t^{mix} before it is passed to the frozen decoder D , mapping it to the
 129 input distribution that D was trained on without modifying D itself. Their joint optimization follows
 130 a single variational rate-distortion objective derived in Sec. 3.3. Notation is summarized in Table 1.
 131 The encoder, decoder, and bitstream interfaces are illustrated in Fig. 1.

132 3.2 Token reuse and residual coding

133 **Reuse predictor.** For token slot i at frame t , the reuse predictor outputs the logit of a Bernoulli
 134 posterior conditioned on the current and reference token features and a temporal context c_i^t :

$$\ell_i^t = f_\phi(z_i^t, \hat{z}_i^{t-1}, c_i^t), \quad q_\phi^p(\rho_i^t = 1 \mid x_t, c_i^t) = \sigma(\ell_i^t), \quad (2)$$

135 where σ is the sigmoid and f_ϕ is a two-layer MLP with GeLU activations. The final-layer bias is
 136 initialized to -0.1 , a near-symmetric prior that avoids both the all-reuse and all-transmit attractors
 137 during the joint warm-up of the reuse predictor and the residual quantizer. To preserve gradients
 138 through the binary decision, training uses the Gumbel-sigmoid relaxation

$$\tilde{\rho}_i^t = \sigma((\ell_i^t + g)/\tau), \quad g \sim \text{Logistic}(0, 1), \quad (3)$$

139 with τ held at 1.0 for a 300-step warm-up and then annealed linearly to 0.5 over the next 2.5×10^3 steps.
 140 The remaining schedule is described in Sec. 3.3. At inference, the hard decision $\rho_i^t = \mathbf{1}[\sigma(\ell_i^t) > 0.25]$
 141 is entropy-coded under the prior of Eq. (9). The resulting train-test consistency is discussed in
 142 Sec. 3.4.

143 **Residual vector quantizer.** For tokens with $\rho_i^t = 0$, we encode the feature-space residual $\delta_i^t =$
 144 $z_i^t - \mathcal{C}[\hat{z}_i^{t-1}]$ through Q stages of vector quantization with per-stage codebook size K_r (we write q
 145 for the stage index, with $1 \leq q \leq Q$):

$$\hat{\delta}_i^t = \sum_{q=1}^Q \text{VQ}_q \left(\delta_i^t - \sum_{q' < q} \text{VQ}_{q'}(\cdot) \right), \quad \Delta_q(r_i^t) \equiv \hat{\delta}_i^t, \quad (4)$$

146 with codebooks updated by exponential moving average and dead codes revived from in-batch
 147 features. The reconstruction in Eq. (1) follows directly. Codebooks are trained jointly with the rest of

the model under the objective in Eq. (8). Gradients do not propagate into $\mathcal{T}$, preserving the perceptual reconstruction floor of the pretrained tokenizer and avoiding the joint encoder-codec instabilities documented for fully co-trained tokenizer-codec systems [Yu et al., 2024a].

The role of residual coding is to exploit the low entropy of δ_i^t relative to z_i^t ; temporal context already explains most of the token feature, so the residual concentrates the remaining variation in a small number of high-energy directions that a modest- depth RVQ can encode in far fewer bits than the raw token index. Without this path, every transmitted token would have to carry its full $\log_2 K$ bit budget, and the routing decision in Eq. (2) would degenerate to a binary skip flag rather than a true rate-distortion trade-off. Ablation variant A4.6 in Sec. 5 quantifies this degradation empirically.

The reference $\hat{z}_i^{t-1}$ is the previous frame’s reconstructed feature at index i . We instantiate the framework with TiTok [Yu et al., 2024b], whose one-dimensional semantic indices identify token slots without spatial correspondence. The formulation transfers to spatially-laid tokenizers without modification. The temporal context c_i^t is built from the current and reference token features at the same slot together with a small temporal-memory cross-attention block whose memory holds the previous W reconstructed frames. The cross-attention is the only non-linear operator in g_ψ . We use $W = 4$ memory frames throughout. Sweeps over $W \in \{1, 2, 4, 8\}$ are reported in Sec. 5.

3.3 Variational RD objective and two-stage multi-rate training

Variational formulation. We view the codec as variational lossy compression with a structured discrete latent $u_t = (\rho_t, r_t)$, where $\rho_t = \{\rho_i^t\}_{i=1}^N$, $r_t = \{r_i^t\}_{i=1}^N$, and the context bundle is $c_t = \{c_i^t\}_{i=1}^N$. The generative model factorizes over the prior and the conditional likelihood,

$$p_\theta(x_t, u_t | u_{<t}) = p(x_t | \hat{z}_t) p_\theta(r_t | \rho_t, c_t) p_\theta(\rho_t | c_t), \quad (5)$$

where $\hat{z}_t$ is the deterministic image of u_t given $u_{<t}$ through Eqs. (1) and (4), and c_t is built from $u_{<t}$ as in Sec. 3.2. The variational posterior factorizes per token and per stage,

$$q_\phi(\rho_t, r_t | x_t, c_t) = \prod_{i=1}^N q_\phi^\rho(\rho_i^t | x_t, c_i^t) q_\phi^r(r_i^t | \rho_i^t, x_t, c_i^t), \quad (6)$$

realized by the reuse predictor of (2) and the deterministic RVQ encoder implicit in (4). The negative evidence lower bound of Eq. (5) under posterior (6) decomposes as

$$-\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_{q_\phi}[-\log p(x_t | \hat{z}_t)]}_{\mathcal{D}} + \underbrace{\text{KL}(q_\phi^\rho || p_\theta(\rho_t | c_t))}_{\mathcal{R}_\rho} + \underbrace{\mathbb{E}_{q_\phi^r}[\text{KL}(q_\phi^r || p_\theta(r_t | \rho_t, c_t))]}_{\mathcal{R}_r}. \quad (7)$$

With the standard variational-compression identification of the reconstruction term as a per-frame distortion and of each KL as the expected coding length under arithmetic coding [Ballé et al., 2018, Minnen et al., 2018], the β -weighted negative ELBO with $\beta = 1/\lambda$ recovers the Lagrangian rate-distortion loss

$$\mathcal{L}_{\text{RD}}(\lambda) = \mathcal{D} + \lambda (\mathcal{R}_\rho + \mathcal{R}_r), \quad (8)$$

which is the loss optimized at every training step. Different rate points correspond to different values of λ , and a model family covering the operating range is constructed by sweeping λ rather than by changing architecture or loss.

This formulation localizes the difference from prior token codecs in a single object: the prior $p_\theta(\rho_t | c_t)$. In TVC the prior is a Dirac on a deterministic checkerboard pattern [Zhou et al., 2025]; in TiTok it is a Dirac on a clip-level partition into time-invariant and time-variant token sets [Chen et al., 2026]. In our framework it is a learned distribution over per-token, per-frame reuse, and the posterior q_ϕ^ρ that minimizes Eq. (7) is content-adaptive at the same granularity. Because $\mathcal{R}_\rho$ is explicitly part of the objective, the indicator stream is entropy-coded under the learned prior, and its rate concentrates near zero wherever the prior is sharp.

Two-stage training. Direct optimization of Eq. (8) from random initialization is unstable: with f_ϕ randomly initialized, q_ϕ^ρ provides no content-discriminative signal, and the marginal posterior over ρ_i^t collapses to a constant before either the reuse predictor or the residual quantizer acquires content sensitivity. We therefore adopt a stabilized schedule with two stages.

190 All training stages use UCF-101 and Kinetics-600 video clips, with random 256×256 crops, clips of
 191 length 7 (treated as 1 I-frame followed by 6 P-frames), and AdamW (learning rate 10^{-4} in Stage 1,
 192 5×10^{-5} in Stage 2). In Stage 1 (single-rate stabilization at fixed $\lambda = \lambda_{\text{mid}}$, with $\lambda_{\text{mid}} = 10^{-2}$ unless
 193 stated otherwise), training proceeds through three sub-phases denoted 1A, 1B, and 1C. The first
 194 5×10^3 iterations force $\rho \equiv 0$ and update only the RVQ codebooks and the residual entropy model,
 195 so codebook utilization stabilizes before reuse is introduced. The next 1.5×10^4 iterations enable the
 196 reuse predictor with two anti-collapse terms,

$$\mathcal{L}_{1B} = \mathcal{L}_{RD}(\lambda_{\text{mid}}) - \beta_H(t) \mathcal{H}(q_\phi^\rho) + \beta_T [\rho^* - \bar{\rho}_t]_+^2,$$

197 where $\bar{\rho}_t$ is the batch-averaged reuse rate and $[\cdot]_+$ is the positive part. The first term, with $\beta_H(0) = 0.7$
 198 decaying linearly to 0 over 2×10^4 steps, encourages a high-entropy posterior so that q_ϕ^ρ neither
 199 saturates nor collapses before content sensitivity emerges. The second term is a one-sided underuse
 200 penalty that activates only when the empirical reuse rate falls below a target ρ^* ; we use $(\rho^*, \beta_T) =$
 201 $(0.08, 2.0)$ in sub-phase 1B and relax to $(0.06, 1.2)$ in 1C. The penalty is essential in our setting:
 202 without it, the residual path is sufficient to drive distortion down on its own, and the reuse posterior
 203 collapses to $\rho \equiv 0$ before the entropy model has learned any temporal structure. The remaining
 204 1.2×10^5 iterations optimize Eq. (8) directly with both auxiliary terms annealed off.

205 In Stage 2 (multi-rate fine-tune), K branches initialize from the Stage 1 checkpoint and fine-tune for
 206 3×10^4 iterations each at distinct rate-distortion targets, parameterised by a sweep of the rate weight λ
 207 over a small set of values that span the operating bitrate range. The exact values used at each evaluation
 208 point are listed in Sec. 4. Branches are independent and parallelizable, and their checkpoints jointly
 209 cover the full reported rate range. The optimizer (AdamW, learning rate 5×10^{-5} with cosine decay to
 210 5×10^{-6}), batch size, and data sampling are unchanged from Stage 1. The distortion term $\mathcal{D}$ in Eq. (8)
 211 is a weighted sum of a feature-space mean-squared error $\|z_t - \hat{z}_t\|_2^2$, which keeps $\hat{z}_t$ on the tokenizer
 212 manifold, and an image-space term $\|x_t - D(\hat{z}_t)\|_2^2 + \alpha_{\text{lpiips}} \text{LPIPS}(x_t, D(\hat{z}_t))$ that is added with
 213 weight 0.1 from sub-phase 1C onward (we set $\alpha_{\text{lpiips}} = 0.5$). Because D is frozen, the gradient of the
 214 image-space term flows through D to $\hat{z}_t$ but not into D itself.

215 3.4 Conditional entropy modeling

216 The priors in Eq. (5) are realized by a conditional entropy model that supplies the coding distributions
 217 for both streams. Letting h_θ^ρ and $\{h_{\theta,q}^r\}_{q=1}^Q$ denote the per-stream heads, the joint prior factorizes
 218 across token positions and across RVQ stages,

$$p_\theta(\rho_t | c_t) = \prod_{i=1}^N \text{Bern}(\sigma(h_\theta^\rho(c_i^t))), \quad (9)$$

$$p_\theta(r_t | \rho_t, c_t) = \prod_{i: \rho_i^t=0} \prod_{q=1}^Q \text{Cat}(\text{softmax } h_{\theta,q}^r(c_i^t, r_i^{t,<q})), \quad (10)$$

219 where h_θ^ρ is a two-layer MLP returning a Bernoulli logit and each $h_{\theta,q}^r$ is a two-layer MLP returning
 220 a categorical logit vector of dimension K_r . Conditioning each stage on its predecessors makes the
 221 residual prior strictly autoregressive across quantization stages. The same MLPs are used at training
 222 time to estimate $\mathcal{R}_\rho$ and $\mathcal{R}_r$ in Eq. (7) and at inference time to drive the arithmetic coder, so the
 223 train-time KL terms in Eq. (7) match the realized bitstream length up to the sub-bit slack of arithmetic
 224 coding.

225 Two architectural choices distinguish this design from the entropy models commonly used in
 226 continuous-latent learned video compression. First, the priors in Eqs. (9)–(10) condition only
 227 on the temporal context c_i^t , which is identical at the encoder and the decoder. The priors do not
 228 condition on the current frame x_t , which is unavailable at the decoder. The variational posteriors of
 229 Eq. (6) do depend on x_t , which is precisely why they are summarized into the bitstream by sampling
 230 ρ and emitting r rather than recomputed at decode time. Second, conditioning on the same c_i^t across
 231 both streams factorizes the bitstream into a reuse stream and a residual stream that can be coded
 232 independently per frame, which keeps the arithmetic coder simple and makes the decoder strictly
 233 causal in time.

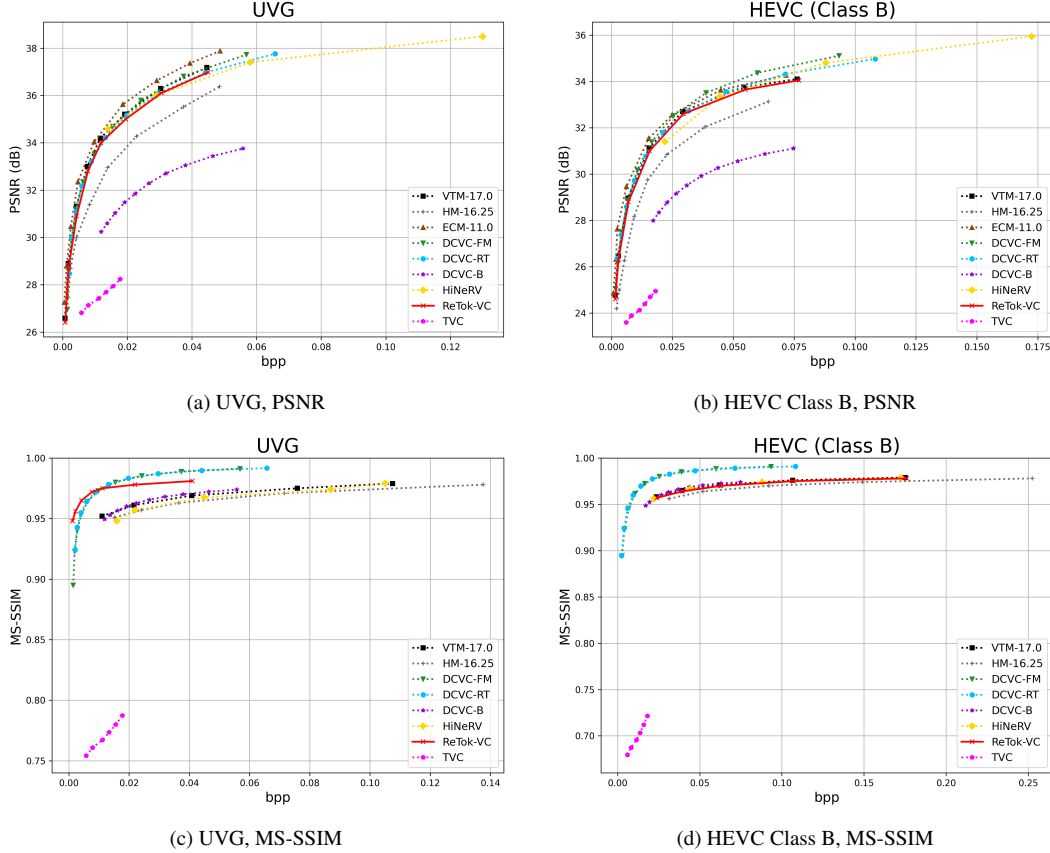


Figure 2: Rate-distortion comparison on UVG (left column) and HEVC Class B (right column) against learned and traditional baselines. Top row: PSNR (dB) versus bits per pixel (bpp), shown for (a) UVG and (b) HEVC Class B. Bottom row: MS-SSIM versus bpp, shown for (c) UVG and (d) HEVC Class B. The solid red line is ReTok-VC; dotted lines are learned (DCVC-FM, DCVC-RT, DCVC-B, HiNeRV) and traditional (VTM-17.0, HM-16.25, ECM-11.0) baselines, plus the token peer TVC.

234 4 Experiments

235 4.1 Implementation Details

236 **Dataset.** Training uses video clips from UCF-101 and Kinetics-600, randomly cropped to 256×256 at
 237 clip length 7 (one I-frame followed by six P-frames, GoP=7). Evaluation is conducted on the standard
 238 UVG and HEVC Class B benchmarks, using the first 100 frames per sequence under matched GoP
 239 cadence. The training and evaluation corpora are fully disjoint.

240 **Implementation details.** ReTok-VC is trained on 8 NVIDIA H800 GPUs. Stage 1 runs 1.4×10^5 total
 241 iterations across three sub-phases at fixed $\lambda = \lambda_{\text{mid}}$. Phase 1A is 5×10^3 iterations of RVQ codebook
 242 warm-up with $\rho \equiv 0$, updating only the residual quantizer and its entropy head. Phase 1B is 1.5×10^4
 243 iterations enabling the reuse predictor under the entropy and underuse anti-collapse regularizers of
 244 Sec. 3.3. Phase 1C is 1.2×10^5 iterations optimizing the full Lagrangian Eq. (8) with both auxiliary
 245 terms annealed off and the image-space term and LPIPS turned on. Stage 2 fine-tunes rate-targeted
 246 branches in parallel from the Stage 1 checkpoint, each for 3×10^4 iterations under a sweep of the rate
 247 weight λ that spans the operating bitrate range. We use AdamW with learning rate 10^{-4} in Stage 1
 248 and 5×10^{-5} with cosine decay to 5×10^{-6} in Stage 2. The frozen image tokenizer is held fixed
 249 throughout. Full hyperparameters are listed in Sec. 3.3 and in the supplementary.

250 **Baselines.** We compare against traditional anchors (VTM-17.0, HM-16.25, ECM-11.0), continuous-
 251 latent learned codecs (DCVC-FM, DCVC-RT, DCVC-B), the implicit-neural-representation codec

HiNeRV, and the token codec TVC [Bross et al., 2021, Sullivan et al., 2012, Li et al., 2024, Jia et al., 2025, Sheng et al., 2025, Kwan et al., 2023, Zhou et al., 2025]. To reduce protocol bias, all methods are evaluated on the same frame budget with matched dataset preprocessing, GoP-level evaluation cadence, and identical metric implementations. Reported points correspond to actually produced bitstreams rather than proxy rate surrogates. Traditional anchors use standard reference configurations.

Evaluation metrics. Distortion is reported as PSNR (RGB) for fidelity and MS-SSIM for structural similarity, against bits per pixel (bpp) measured from the produced bitstream. Aggregate comparisons use BD-rate computed in two windows: an *ultra-low* window (0.005–0.028 bpp) for strict head-to-head with token peers in their reported regime, and a *full* window (0.005–0.50 bpp) for broader operating behavior. The split prevents over-claiming outside a peer’s reported bitrate range while preserving a unified RD narrative for ReTok-VC.

4.2 Main results

Fig. 2 reports rate-distortion curves on UVG and HEVC Class B in PSNR and MS-SSIM. We compare ReTok-VC against four baseline families: traditional codecs (VTM-17.0, HM-16.25, ECM-11.0), continuous-latent learned codecs (DCVC-FM, DCVC-RT, DCVC-B), the implicit-neural-representation codec HiNeRV, and the token peer TVC.

UVG, PSNR (Fig. 2a). ReTok-VC tracks VTM-17.0 and HM-16.25 within roughly 0.2 dB in the overlap window, and stays within 0.5 dB of the strongest DCVC variant (DCVC-FM) across the same range. ECM-11.0’s edge at high bpp reflects decades of engineering on block-hybrid mode decisions and quadtree partitioning, which become decisive once enough bits are available. Within the DCVC family, DCVC-RT trades roughly 0.2 dB for real-time decoding, and DCVC-B is confined to the higher-bpp end because its B-frame design assumes bidirectional context. HiNeRV begins at 0.014 bpp and crosses ReTok-VC in the mid-bpp regime, since INR codecs gain quality only once the network has enough capacity to overfit a single sequence. TVC sits 4 to 7 dB below ReTok-VC across the bitrate window where the two curves overlap, and terminates near 0.018 bpp because its checkerboard mask transmits a fixed half of the tokens regardless of content.

HEVC Class B, PSNR (Fig. 2b). The same ordering is preserved. ECM-11.0 again leads at high bpp because Class B sequences are dominated by camera motion and detailed texture, the regime in which next-generation block partitioning matters most. The DCVC family is tighter on Class B than on UVG because its training corpus is closer to the Class B resolution and motion statistics, and the gap between DCVC-FM and DCVC-RT shrinks to under 0.3 dB. ReTok-VC remains within the VTM and HM envelope and within 0.5 dB of DCVC-FM at matched bpp. TVC operates only in 0.006 to 0.018 bpp and falls 4 to 7 dB below ReTok-VC inside that window. The gap widens at the upper end because fixed-mask routing cannot keep up with the higher motion magnitude.

UVG, MS-SSIM (Fig. 2c). All continuous-latent and traditional baselines cluster in a narrow band, compressing to 0.97–0.99 at moderate bpp. ReTok-VC sits inside the cluster, indicating that the discrete-token reuse path carries no structural cost relative to continuous-latent codecs. TVC is the lone outlier at 0.75 to 0.79. The penalty stems from its deterministic checkerboard masking: the same spatial half of tokens is dropped in every frame regardless of motion, so any persistent texture in those positions is corrupted in every reconstructed frame. ReTok-VC’s content-adaptive reuse posterior retains unchanged tokens and only spends bits where content has moved, which is the structural-fidelity invariant that MS-SSIM rewards.

HEVC Class B, MS-SSIM (Fig. 2d). The non-TVC cluster shifts to 0.93–0.99 because Class B’s higher motion taxes structural similarity at every operating point. ReTok-VC again falls inside this cluster, while TVC sits at 0.68 to 0.72, an even larger gap than on UVG. A static-looking patch in one frame may belong to a moving object in the next, so dropping a fixed half of the tokens consistently corrupts spatial structure on the harder content. ReTok-VC’s per-token, per-frame reuse decision conditions on local temporal context, which is why our curve does not degrade.

5 Ablation

We use the training schedule itself as an ablation. Each successive phase is initialized from the preceding checkpoint and activates exactly one additional component from Sec. 3.3, so row-to-row

Table 2: Phase-progression ablation at $\lambda \in \{10^{-2}, 10^{-3}\}$, ablating each successive training-phase component.

λ	Metric	1A (RVQ)	1B (+Reuse)	1C (+Perc.+Img.)	Stage 2 (+Sched.)
10^{-2}	Loss	0.728	0.380	0.401	0.389
	Token reuse	0.0000	0.3459	0.3340	0.3385
	BPP	0.0618	0.0366	0.0374	0.0369
	PSNR	31.69	31.037	32.587	33.961
10^{-3}	Loss	0.361	0.096	0.118	0.109
	Token reuse	0.0000	0.3218	0.3015	0.2978
	BPP	0.0658	0.0408	0.0415	0.0410
	PSNR	32.43	31.352	32.968	33.356

deltas in Table 2 isolate the marginal effect of (b) entropy-coded learned reuse, (c) the perceptual and image-space distortion terms, and (d) the Stage-2 multi-rate fine-tuning schedule. Phase 1A serves as the RVQ-only reference. The reported quantities are token reuse (mean hard reuse ratio), BPP (bits per pixel), and PSNR computed from the mean image MSE as $10 \log_{10}(1/\text{MSE})$. We discuss each transition in turn.

Effect of learned reuse (1A $\rightarrow$ 1B). Activating the reuse predictor with the entropy and underuse anti-collapse terms reduces BPP from 0.062 to 0.037 at $\lambda=10^{-2}$, a 40.8% reduction, and from 0.066 to 0.041 at $\lambda=10^{-3}$, a 38.0% reduction. The hard reuse ratio rises from 0 to 0.346 and 0.322 respectively, confirming that the underuse penalty prevents the all-transmit collapse anticipated in Sec. 3.3. PSNR temporarily drops by 0.65 and 1.08 dB, which is expected because the residual quantizer has to absorb all distortion on the non-reused tokens at unchanged distortion weight. The fidelity loss is recovered in the next phase.

Effect of perceptual and image-space distortion (1B $\rightarrow$ 1C). Adding the LPIPS and image-space MSE terms with weight 0.1 raises PSNR by 1.55 dB at $\lambda=10^{-2}$ and by 1.62 dB at $\lambda=10^{-3}$, while BPP increases by under 2.5% in both regimes. Reuse ratio stays within 0.012 of its phase-1B value, indicating that the gain comes from refining the residual reconstruction rather than from re-routing tokens. Stage 1C therefore closes the fidelity gap opened during reuse activation at a negligible rate cost.

Effect of multi-rate fine-tuning (1C $\rightarrow$ Stage 2). The cosine-annealed Stage-2 schedule contributes an additional 1.37 dB at $\lambda=10^{-2}$ and 0.39 dB at $\lambda=10^{-3}$, while BPP decreases by 1.3% and 1.2% respectively. Reuse ratio remains stable, so this stage acts as a polish step on top of an already-converged routing.

Effect of λ on rate. At the end of Stage 2, $\lambda=10^{-2}$ yields BPP 0.0369 at PSNR 33.96, and $\lambda=10^{-3}$ yields BPP 0.0410 at PSNR 33.36. The two operating points lie at distinct rates as expected from the variational rate-distortion formulation, so a small sweep of λ in Stage 2 produces a model family that traces out the rate axis without further architectural changes.

6 Conclusion

ReTok-VC frames the per-token, per-frame decision to reuse temporal context as the variational posterior of a structured discrete latent-variable model. Under this view the Lagrangian rate-distortion loss is recovered as the β -weighted negative ELBO, the reuse indicator is entropy-coded under a learned prior conditioned on temporal context, and the residual quantizer refines transmitted tokens in feature space. The framework operates on top of a frozen image tokenizer, and a two-stage schedule yields a single model family that covers the full target rate range without architectural changes.

The present formulation has several limitations. The reconstruction quality of ReTok-VC is bounded by the underlying image tokenizer. The temporal context summarizes only a fixed window of previous reconstructions, so very long-range dependencies are not explicitly modeled. The model also assumes a stable per-slot token identity, which restricts the choice of tokenizer to spatially-aligned or semantic-slot designs.

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